Shakiba Davari

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EDUCATION

Ph.D., Department of Computer Science at Virginia Tech – Specialization: Mixed Method UX & 3D Interaction Design (August 2024)
Dissertation Title: Intelligent Augmented Reality (iAR): Context-aware Inference and Adaptation in AR [4]

M.Sc., Department of Computer Science at Virginia Tech – Specialization: Human-Computer Interaction (August 2020)

M.Sc., Department of Computer Science at Georgia Tech – Specialization: Computational Perception and Robotics

B.Sc., Department of Computer Engineering at Beheshti University, Iran – Major: Computer Hardware (May 2014)

INDUSTRY EXPERIENCE

Research Scientist Intern, Adobe Research

May 2022 – August 2022

- Prototyped 16 iOS interfaces for document navigation in AR and evaluated their performance across varied real-world settings.
- Collected and analyzed survey data from 12 participants and formulated the findings as context-aware design recommendations.

Mixed-Methods UX Intern, Microsoft Research (MSR)

May 2021 – August 2021

- Prototyped an assistive VR interface with customizable virtual monitors for low-vision users.
- Collected qualitative and quantitative data from 21 low-vision participants, synthesized the data, and presented the outcome as actionable design recommendations for future hardware and software improvements to cross-functional teams [6].

RESEARCH EXPERIENCE & ACADEMIC POSITIONS

Postdoctoral Research Fellow, Georgia Tech

January 2025 – present

- Spearhead the writing of an NSF grant proposal on leveraging AR for intuitive human-robot collaboration.
- Lead a team of researchers and developers on the development of an XR tool for collaboration and training.
- Mentor four graduate students to formulate and conduct comprehensive user studies, quantify physiological and performance measurements, and evaluate user experience, usability, and efficiency across robotics, AI, and human factors. Selected projects:
    - Bidirectional AR interactions for human-robot collaboration [1], [2].
    - Personalized AR interaction based on user expertise and contextual behavior.
    - Constructing XR simulations to analyze perceptual experiences of visually impaired individuals, to identify key accessibility barriers and enhance inclusive product design [3].

Graduate Research Assistant, Virginia Tech

August 2018 – August 2024

Supervisor: Doug A. Bowman

- Conducted end-to-end prototyping and mixed-methods evaluation of intelligent XR systems, including concept development, implementation, user studies, quantitative and qualitative data analysis, publication, and presentation of findings and design recommendations for adaptive 3D interfaces and interactions. Selected projects:
    - Adaptive Placement: Context-aware spatial layout in varied social settings and user-mobility contexts [8].
    - Glanceable & Occlusion-aware AR: Lightweight overlays adapted to user attention and occlusion [11], [12].
    - Socially Aware AR: leveraging speech content & computer vision to improve user awareness and efficiency [10].
    - Collaborative 3D Interaction: Techniques to improve presence and collaboration in VR environments [4].
- Developed foundational theoretical frameworks and taxonomies for intelligent XR, including a taxonomy of context, iAR architecture, XR design space, input modality evaluation framework, and classification of AR interfaces [5], [6], [8], [12].

Undergraduate Research Assistant, Center for Automotive Research at Stanford (CARS)

October 2017 – May 2018

- Developed simulation environments to model and analyze the effects of human driving behavior on autonomous vehicle planning algorithms.

Undergraduate Research Assistant, University of Toronto

January 2015 – May 2016

- Automated web crawling and data extraction to inform the design of conversational robots for an online assistive technology rating platform supporting Alzheimer's caregivers.
- Developed machine learning and computer vision algorithms for UAV-based monitoring of indoor construction sites, enabling real-time tracking of construction progress and automated updates to 4D BIM models [13], [14].

SELECTED PROFESSIONAL ENGAGEMENTS

Invited Talks

- Intelligent Augmented Reality (iAR) – *Georgia Tech (March 2025)*
- Towards iAR: Designing Effective AR through Context-Awareness – *Coburg University (October 2024)*
- Context-Aware Inference and Adaptation in iAR – *IEEE ISMAR Workshop on Perceptual and Cognitive Issues in XR (October 2022)*

Awards

- NextProf Nexus Future Faculty Workshop Award – *UC Berkeley (September 2025)*
- Winning Team for Best 3D User Interface Award for Two Consecutive Years – *IEEE VR (March 2021 & March 2022)*
- Graduate Student Service Award – *Department of Computer Science at Virginia Tech (May 2020)*

Academic Community Service

- International Program Committee (IPC) – *IEEE VR Conference 2026*
- Conference Session Co-organizer and Chair: "To Automate or To Augment?" – *IEEE International Conference on Automation Science and Engineering (2025)*
- International Program Committee (IPC) – *IEEE ISMAR Conference 2025*
- Conference Committee Member Poster Session – *IEEE VR Conference 2025*
- Co-organizer: "1st Workshop on intelligent XR (iXR)" – *IEEE ISMAR Conference 2024*
- Conference Session Chair – *ACM Spatial User Interaction 2024*
- Peer Reviewer – *IEEE VR, ISMAR, TVCG and ACM CHI, UIST, AutomotiveUI (2020 – present)*

Mentoring – Akhil Ajikumar, Parisa Ghasemi, Alexander Giovannelli, Steven Yoo, Daniel Stover *(2020 – present)*

Leadership – CS Graduate Student Council & Iranian Society Virginia Tech *(August 2018 – August 2021)*

TOOLS & SKILLS

- C#, Python, Swift, C/C++, Unity3D, MRTK, ARKit, ARCore, Adobe Aero, Photon Networking, JMP, SPSS, R, OpenCV, Pymunck, TensorFlow, PyTorch, Flask, Selenium, Apache Nutch, SimVista, SimCreator
- Quantitative & Qualitative Analysis, Design Guidelines, Context-Aware Design, 3D Interaction, User Studies, Eye Tracking, UX & UI, Mixed-Methods Research, Academic Writing, Interdisciplinary Collaboration and Communication

SELECTED PUBLICATIONS

1. **S. Davari** et al., "Context-Aware Augmented Reality for Human-Robot Collaboration", IEEE ISMAR-Adjunct, 2025
2. A. Ajikumar, **S. Davari**, et al., "Multimodal AR System for Bidirectional Intent Communication in Human-Robot Collaboration", ACM THRI, [Under Review]
3. P. Ghasemi, **S. Davari**, et al., "Empathic Need Finding: Investigating the Impact of Simulated Visual Impairments on Design", [In preparation]
4. A. Giovannelli, L. Pavanatto, **S. Davari**, et al., "Investigating the Influence of Playback Interactivity During Guided Tours for Asynchronous Collaboration in Virtual Reality", IEEE VR, 2022, Pp. 23–33, 2025.
5. **S. Davari** and D. A. Bowman, "Evaluating Input Modalities in AR: A Framework and A Survey on Eye Input" [In Preparation], 2025.
6. **S. Davari** et al., "Towards Intelligent AR (IAR): A Taxonomy of Context, an IAR Architecture, and an Empirical Study", IEEE TVCG [Under Review], 2024.
7. **S. Davari**, "Intelligent Augmented Reality (IAR): Context-Aware Inference and Adaptation in AR", Ph.D. Dissertation, Virginia Tech, 2024.
8. **S. Davari** and D. A. Bowman, "Towards Context-Aware Adaptation in XR: A Design Space for XR Interfaces and an Adaptive Placement Strategy", IEEE TVCG [Under Review], 2024.
9. L. Pavanatto, **S. Davari**, et al., "Virtual Monitors vs. Physical Monitors: an Empirical Comparison for Productivity Work", Frontiers in VR, Vol. 4, 2023.
10. **S. Davari** et al., "Validating the Benefits of Glanceable and Context-Aware AR for Everyday Information Access Tasks", IEEE VR, 2022, Pp. 336–444.
11. F. Lu, **S. Davari**, et al., "Glanceable AR: Evaluating Information Access Methods for Head-Worn AR," In IEEE VR, Mar. 2020, Pp. 930–939.
12. **S. Davari** et al., "Occlusion Management Techniques for Everyday Glanceable AR Interfaces," in IEEE VR Workshops (VRW), 2020, Pp. 324–330.
13. H. Hamledari, **S. Davari**, et al., "UAV-enabled Site-to-BIM Automation: Aerial Robotic- and Computer Vision-Based Development of As-Built/As-Is BIMs and Quality Control," Construction Research Congress 2018, Pp. 336–346, 2018.
14. H. Hamledari, **S. Davari**, et al., "UAV Mission Planning Using Swarm Intelligence and 4D BIMs in Support of Vision-Based Construction Progress Monitoring and As-Built Modeling," Construction Research Congress 2018, Pp. 43–53, 2018. Doi: Https://Doi.Org/10.1061/9780784481264.005.